Heinrich Weber — *Textbook of Algebra*
Working English Translation Batch 01

Translated from the Kimi-cleaned source lane

Preface to the Second Volume

In accordance with the intention announced in the preface to the first volume, I am now able to place the second volume of my *Textbook of Algebra* before the public. The plan laid down there has been carried out in its essential points. In the applications I have tried to select problems that had already acquired an independent interest in other fields, such as geometry or the theory of functions, and that at the same time bring the principal points of algebraic theory before the reader in as many different ways as possible.

The application of the theory of algebraic numbers has been carried through as far as the theory of cyclotomic numbers. If life and strength endure, I hope in a continuation of my work to present the further applications to the domain of elliptic functions, which are contained only in part in my book *Elliptic Functions and Algebraic Numbers.*

During the preparation and printing of the second volume as well, I have had beside me the help and counsel of the friends whom I already named in the preface to the first edition. But the first volume has by now also won for itself many new friends who have advanced my work by suggestions and advice. To them all I express my thanks here, and add the request that they may continue to preserve their interest in the work.

Strasbourg, July 1896.
The Author.

Part I. General Group Theory

§1. Definition of Groups

In the first volume, in connection with permutations, we became acquainted with the concept of a group and made important algebraic applications of it. Our next task must therefore be to formulate more generally this concept, so extraordinarily important in all of modern mathematics, and to become familiar with the laws that govern it. We place the following definition at the head of the discussion.

A system P of objects (elements) of any kind whatsoever becomes a *group* when the following assumptions are satisfied.

1. A rule is given according to which, from a first and a second element of the system, a completely determined third element of the same system is derived.

Symbolically, if a is the first, b the second, and c the third element, one writes

$$ab = c, \qquad c = ab,$$

and says that c is *composed* from a and b, while a and b are the *components* of c.

For this composition one does not in general assume the commutative law; that is, ab may differ from ba. On the other hand, one does assume

2. the *associative law*,

that is, whenever a, b, c are any three elements of P, one has

$$(ab)c = a(bc).$$

From this, by complete induction, it follows that one always arrives at the same result when, in any finite sequence of elements of P, a, b, c, d, $\ldots$, one first composes two neighboring elements, then again two neighboring elements, and so on until the whole sequence has been reduced to a single element, denoted by $abcd\ldots$. Thus, for example,

$$\begin{aligned} abcd &= (ab)cd = [(ab)c]d = (ab)(cd) \\ &= a(bc)d = [a(bc)]d = a[(bc)d] \\ &= ab(cd) = (ab)(cd) = a[b(cd)]. \end{aligned}$$

This file is a working translation artifact only. It translates the cleaned Kimi source opening material and the start of §1; it does not yet claim chapter-complete coverage.